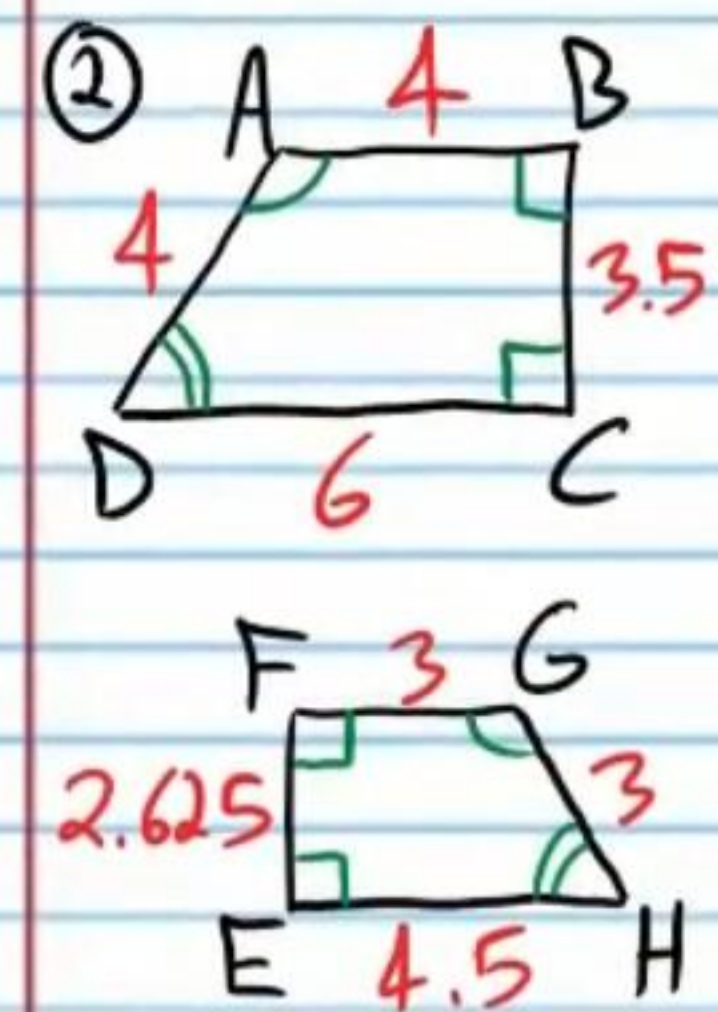
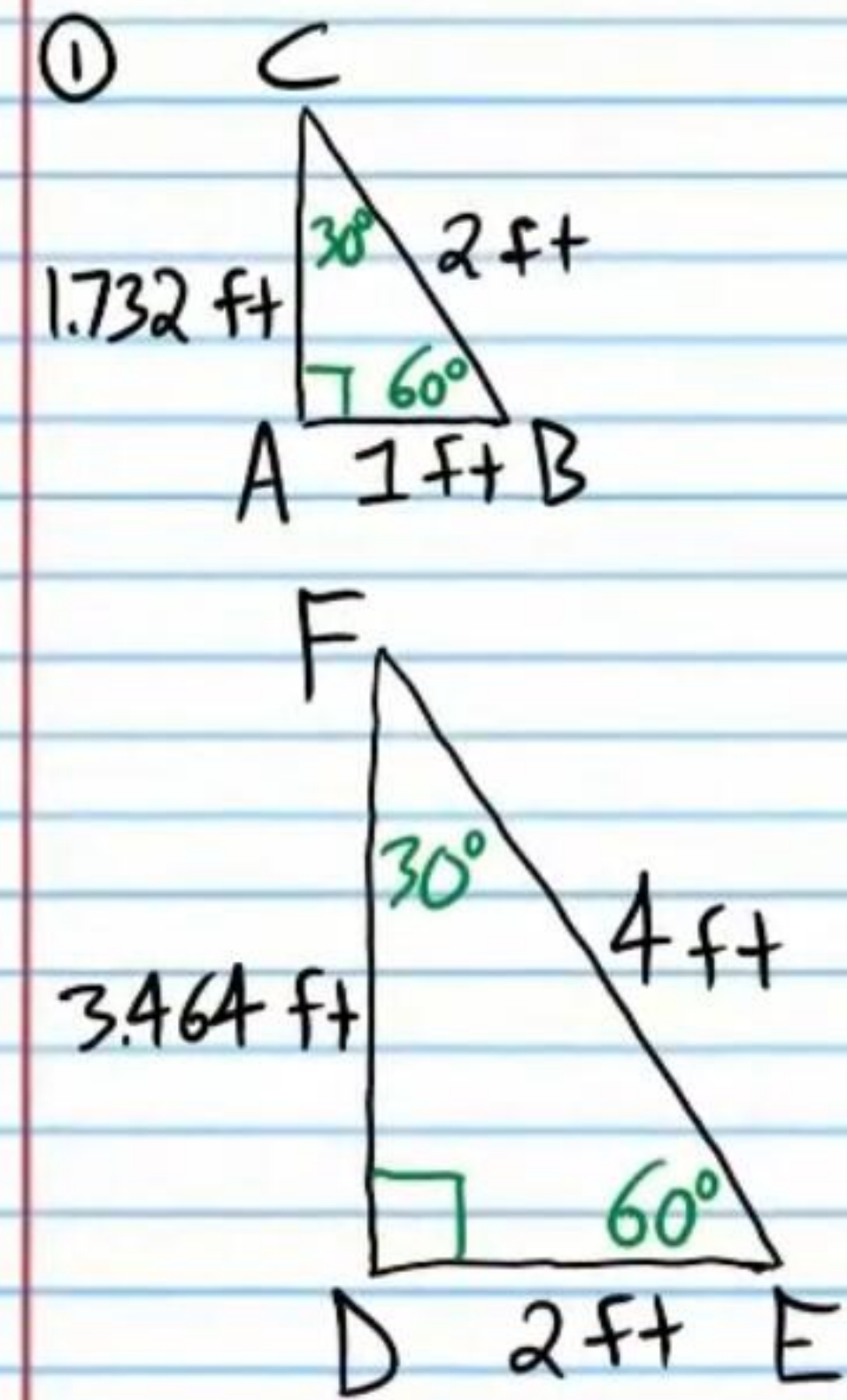


Similar Polygons and Similarity Postulates and Theorems (Homework)

Show that the following polygons are similar. Then, write similarity statements.



Answers

I) $\angle A \cong \angle D, \angle B \cong \angle E$

$\angle C \cong \angle F \checkmark$

II) $\frac{1}{2} = \frac{2}{4} = \frac{1.732}{3.464}$

$\frac{1}{2} = \frac{1}{2} = \frac{1}{2}$

$\frac{AB}{DE} = \frac{CB}{FE} = \frac{CA}{FD} \checkmark$

III) $\triangle ABC \sim \triangle DEF$

I) $\angle A \cong \angle G, \angle D \cong \angle H$
 $\angle B \cong \angle F, \angle C \cong \angle E \checkmark$

II) $\frac{4}{3} = \frac{4}{3} = \frac{6}{4.5} = \frac{3.5}{2.625}$

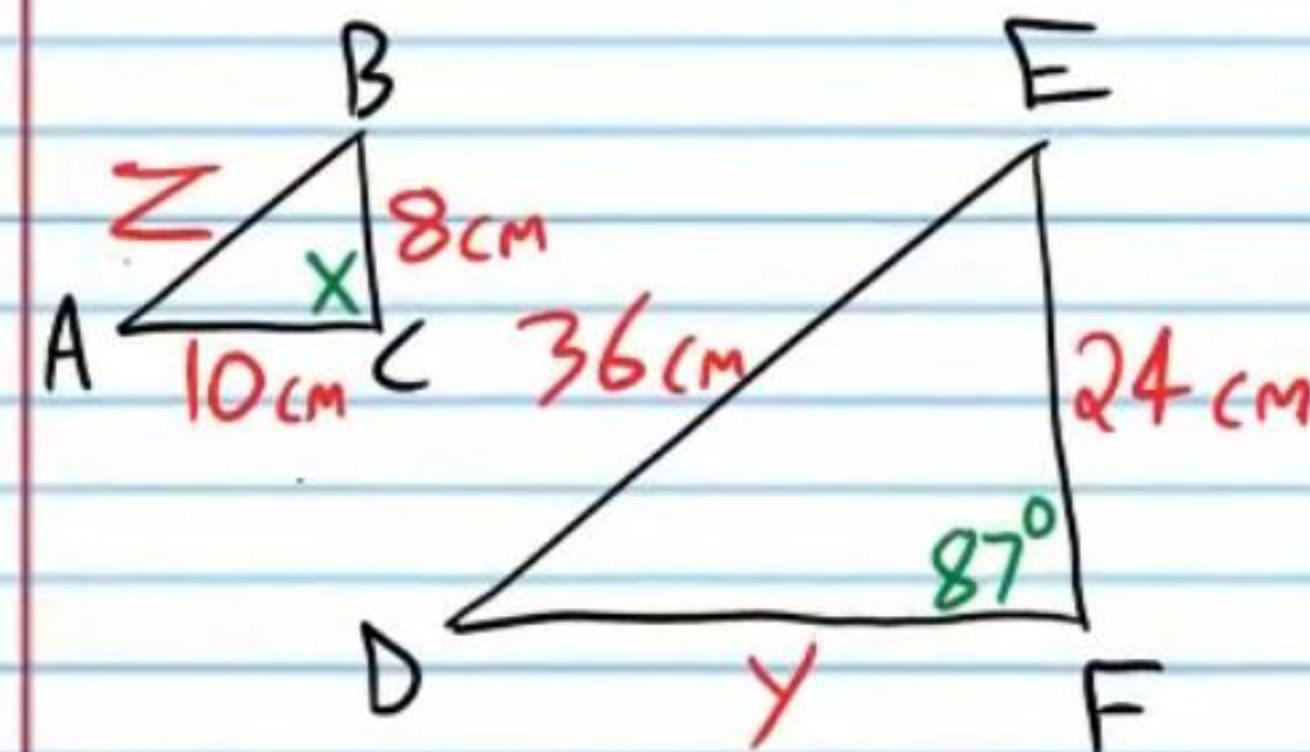
$1.\bar{3} = 1.\bar{3} = 1.\bar{3} = 1.\bar{3}$

$\frac{AB}{FG} = \frac{AD}{GH} = \frac{DC}{EH} = \frac{BC}{FE} \checkmark$

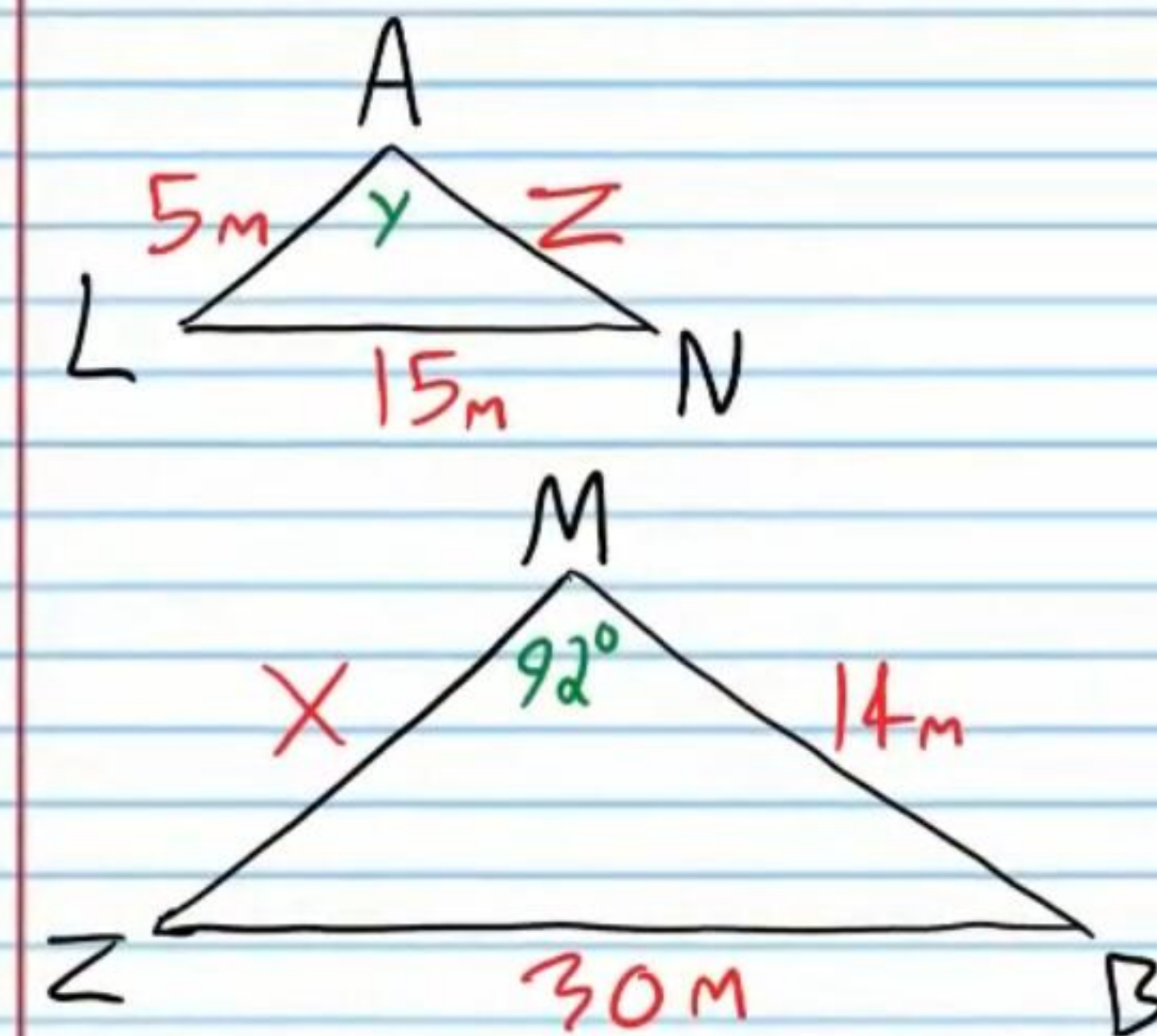
III) $ABCD \sim GFHE$

For each problem below, the polygons in each pair are similar. Find the scale factor and the unknowns.

③ $\triangle ABC \sim \triangle DEF$



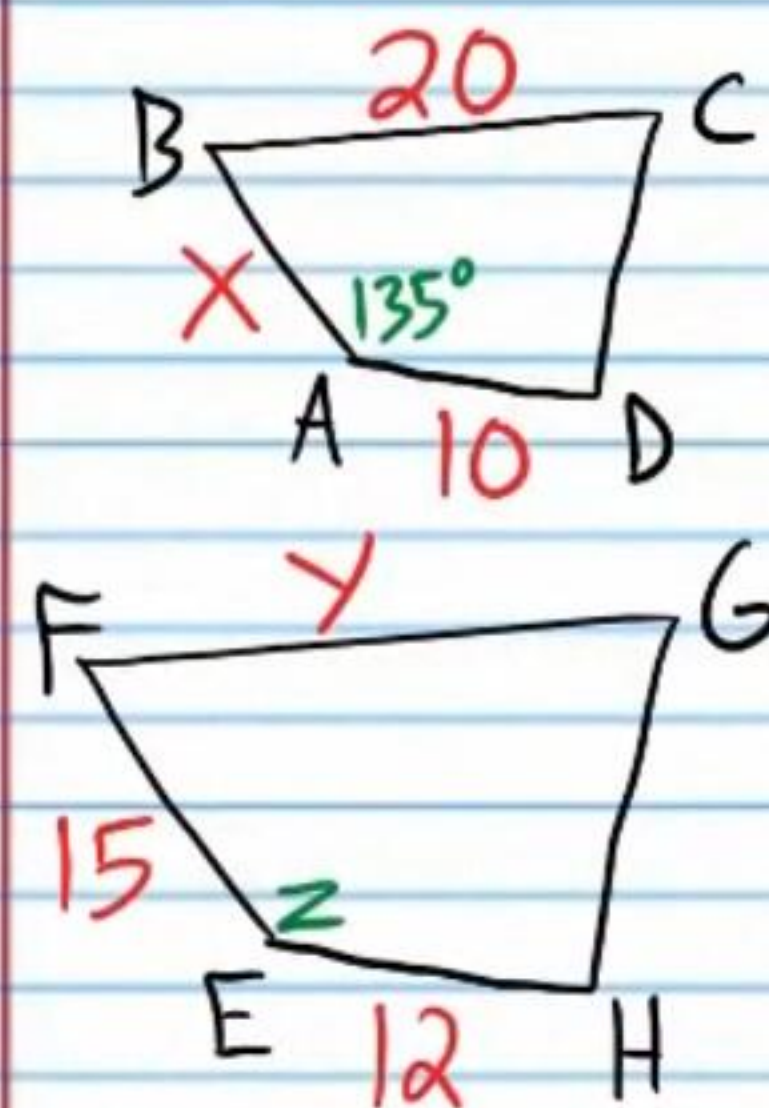
④ $\triangle LNA \sim \triangle ZBM$



$x = 87^\circ$
 $y = 30 \text{ cm}$
 $z = 12 \text{ cm}$
 $SF = 3$
or
 $SF = \frac{1}{3}$

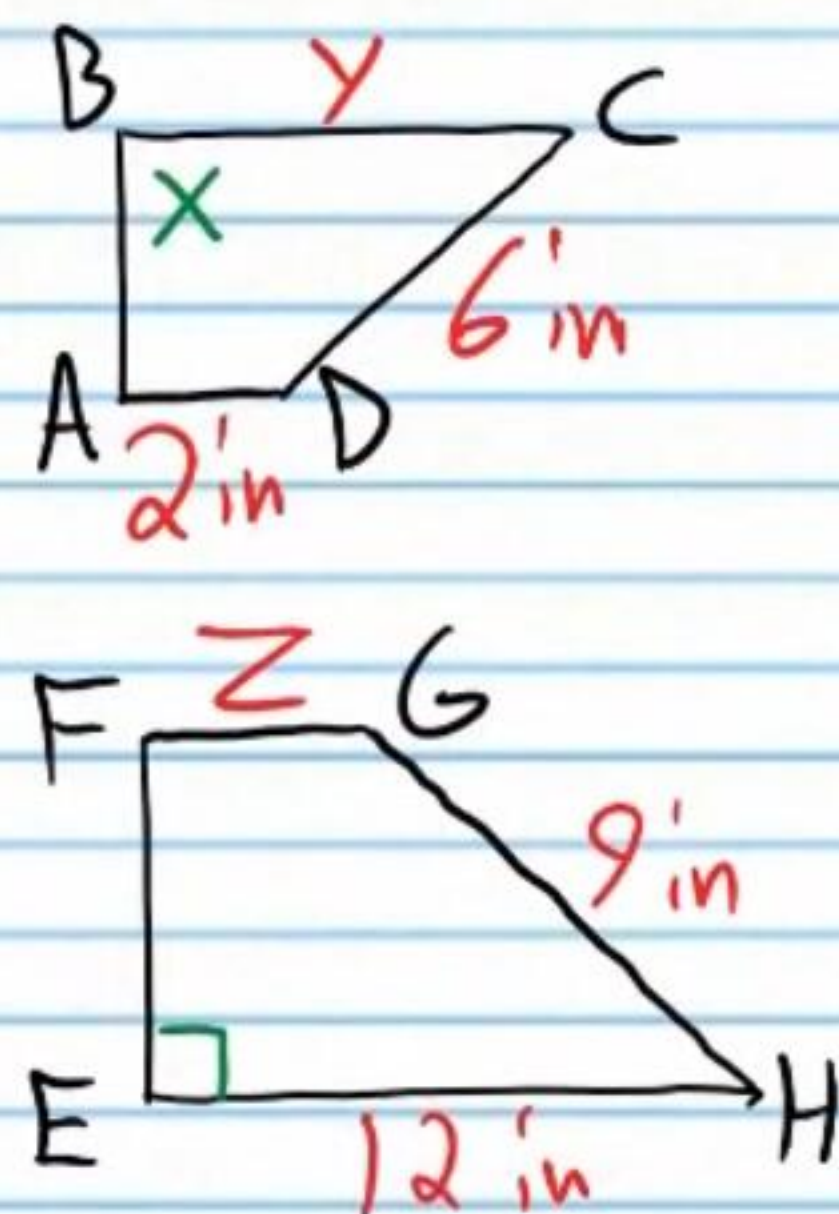
$x = 10 \text{ m}$
 $y = 92^\circ$
 $z = 7 \text{ m}$
 $SF = 2$
or
 $SF = \frac{1}{2}$

⑤ $ABCD \sim EFGH$



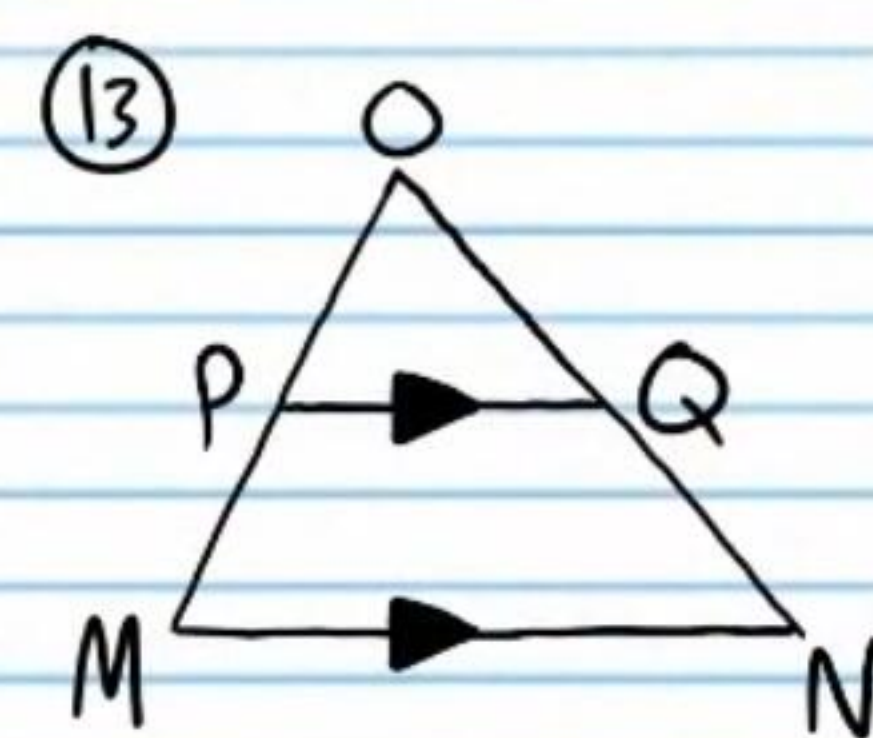
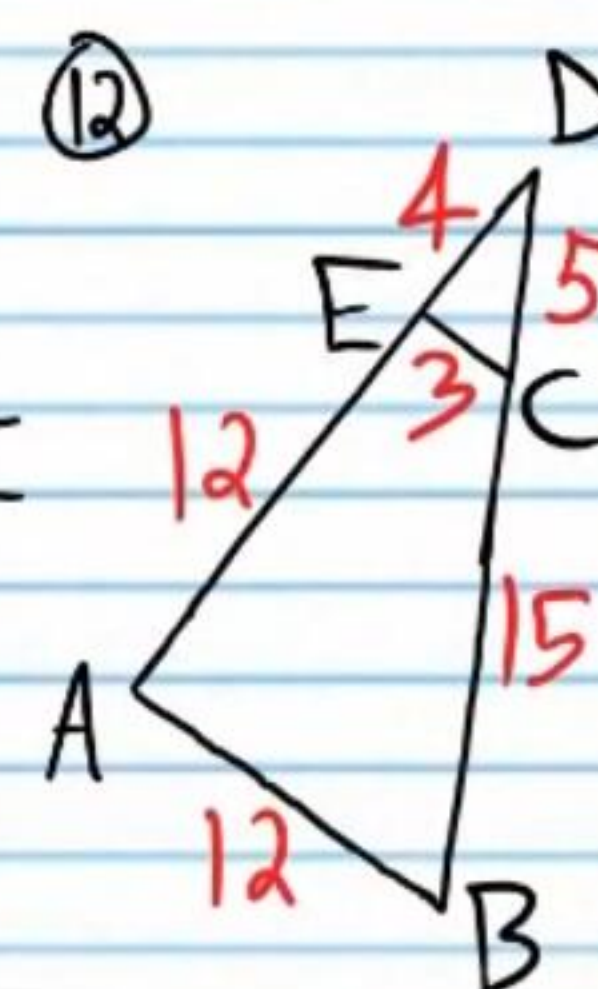
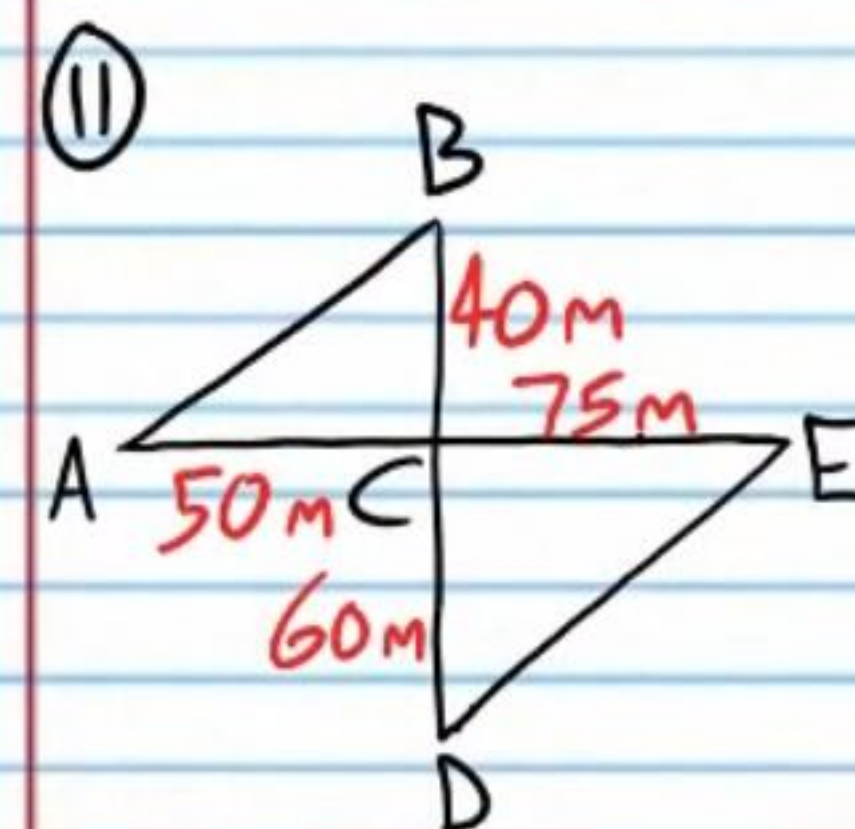
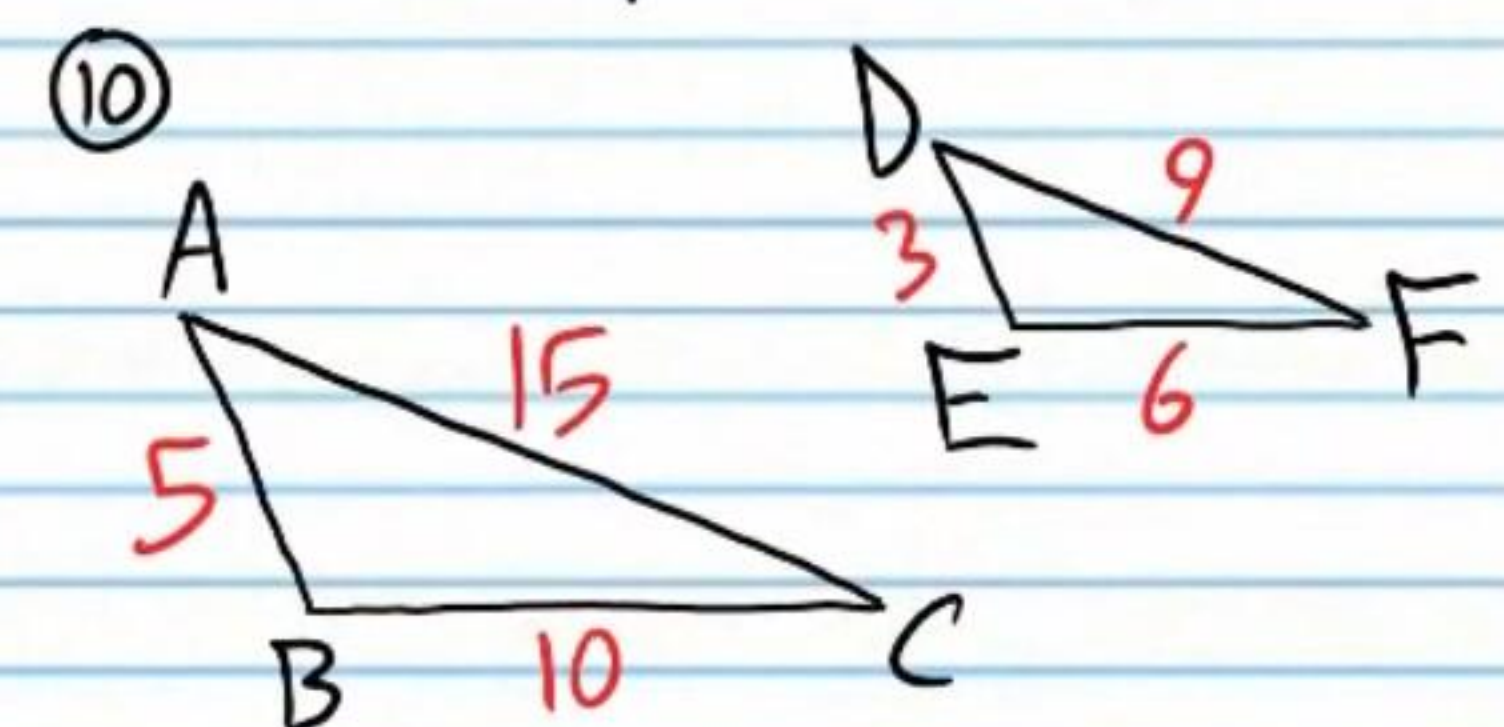
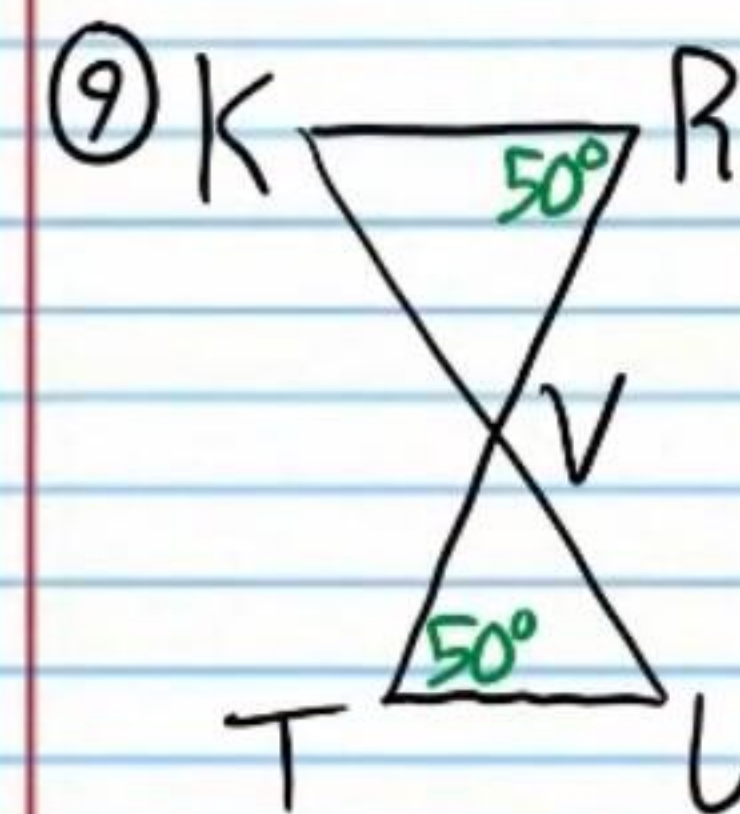
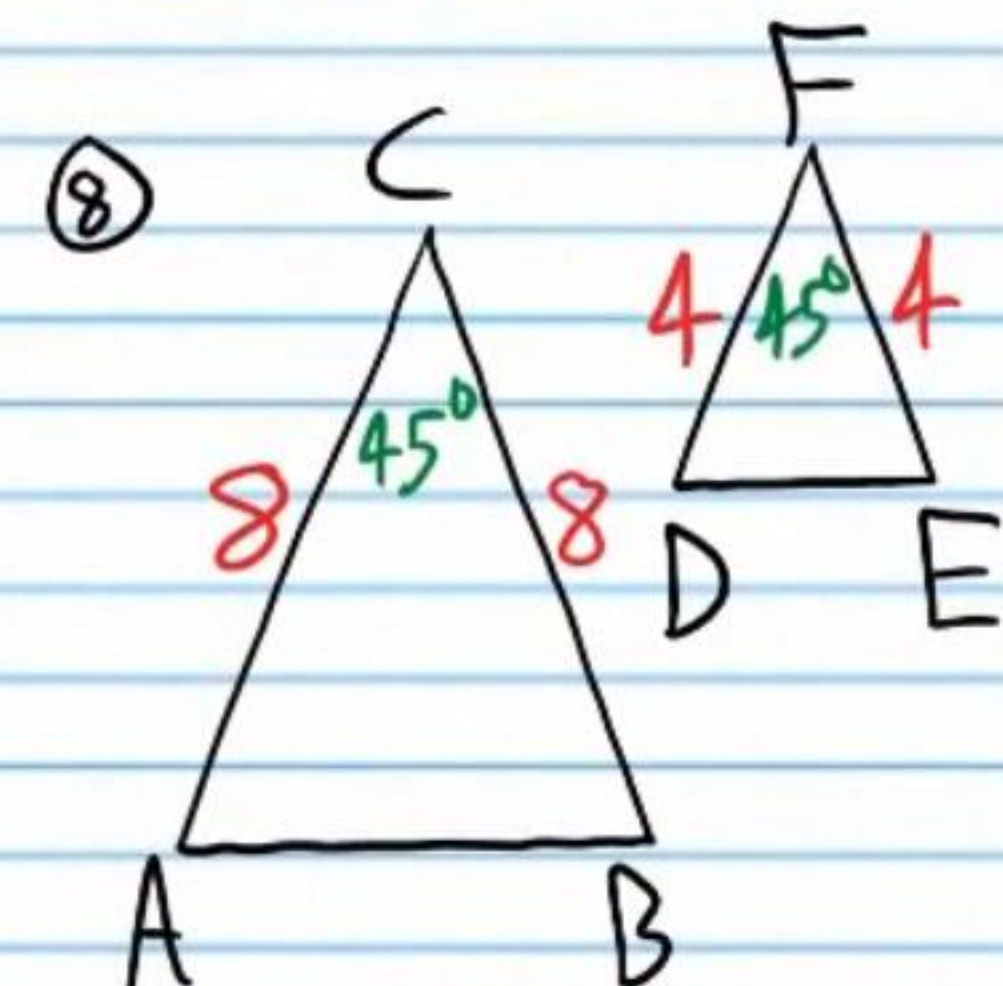
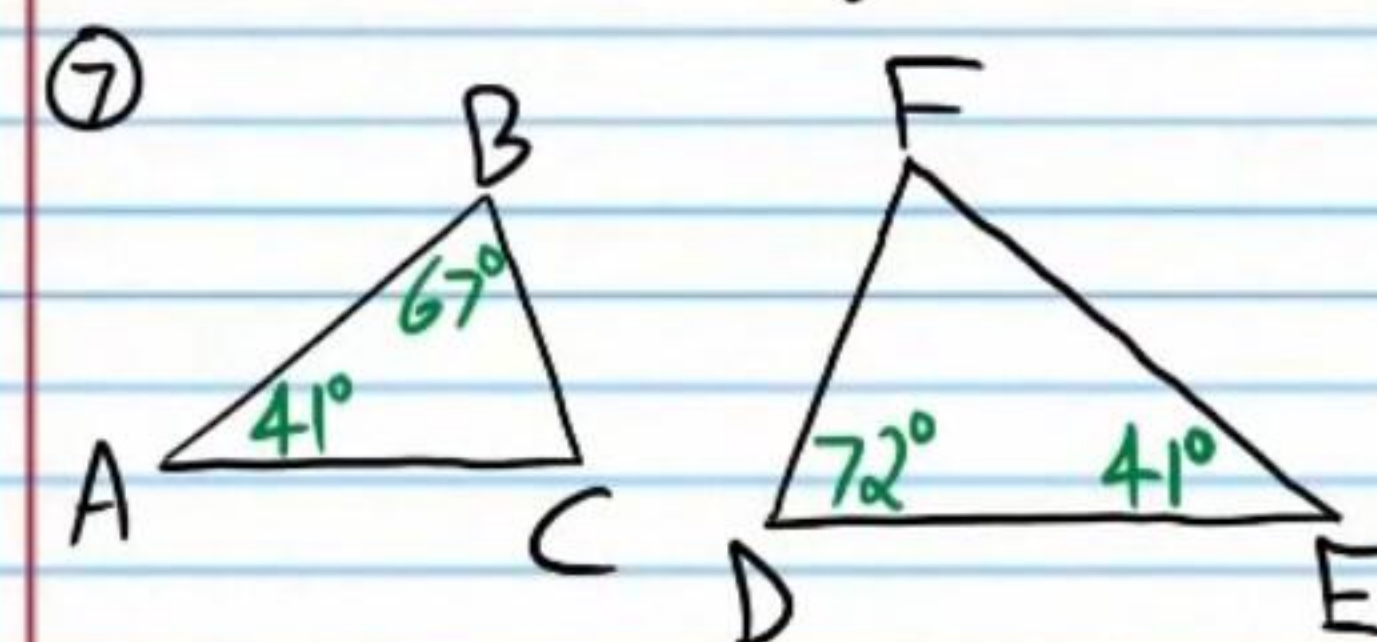
$$\begin{aligned} X &= 12.5 \\ Y &= 24 \\ Z &= 135^\circ \\ SF &= 6/5 \\ \text{or} \\ SF &= 5/6 \end{aligned}$$

⑥ $ABCD \sim FEHG$



$$\begin{aligned} X &= 90^\circ \\ Y &= 8 \text{ in} \\ Z &= 3 \text{ in} \\ SF &= 3/2 \\ \text{or} \\ SF &= 2/3 \end{aligned}$$

In the following problems, write two-column proofs showing that the triangles are similar (answers are on a different page).



Answers for Problem #'s 7-13.

⑦ Statement	Reason
I) $m\angle A = 41^\circ$, $m\angle B = 67^\circ$, $m\angle E = 41^\circ$, and $m\angle D = 72^\circ$	Given (diagram)
II) $m\angle A + m\angle B + m\angle C = 180^\circ$	Δ Sum Thm
III) $41 + 67 + m\angle C = 180^\circ$	Subs. Prop of =.
IV) $108 + m\angle C = 180$	Simplification
V) $m\angle C = 72^\circ$	Subt. Prop of =.
VI) $m\angle A = m\angle E$ and $m\angle D = m\angle C$	Trans. Prop of =.
VII) $\Delta ABC \sim \Delta EFD$	AA

⑧ Statement	Reason
I) $m\angle C = 45^\circ$, $m\angle F = 45^\circ$, $AC = 8$, $BC = 8$, $DF = 4$, and $FE = 4$.	Given (diagram).
II) $m\angle C = m\angle F$	Trans. Prop of =.
III) $\frac{AC}{DF} = \frac{8}{4}$, $\frac{BC}{FE} = \frac{8}{4}$	Subs. Prop of =.
IV) $\frac{AC}{DF} = 2$, $\frac{BC}{FE} = 2$	Simplification.

V) $\frac{AC}{DF} = \frac{BC}{FE}$	Trans. Prop of =.
VI) $\Delta ABC \sim \Delta DEF$	SAS

⑨ Statement	Reason
I) $m\angle R = 50^\circ$ and $m\angle T = 50^\circ$	Given (diagram)
II) $m\angle R = m\angle T$	Trans. Prop of =.
III) $m\angle KVR = m\angle TVU$	Vert. $\angle$'s Thm
IV) $\Delta KRV \sim \Delta UTV$	AA

⑩ Statement	Reason
I) $AB = 5$, $AC = 15$, $BC = 10$, $DE = 3$, $DF = 9$, and $EF = 6$.	Given (diagram)
II) $\frac{AB}{DE} = \frac{5}{3}$, $\frac{AC}{DF} = \frac{15}{9}$, and $\frac{BC}{EF} = \frac{10}{6}$	Subs. Prop of =.
III) $\frac{AB}{DE} = \frac{5}{3}$, $\frac{AC}{DF} = \frac{5}{3}$, and $\frac{BC}{EF} = \frac{5}{3}$	Simplification.
IV) $\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$	Trans. Prop of =.
V) $\Delta ABC \sim \Delta DEF$	SSS

⑪ Statement	Reason
I) $AC = 50\text{m}, BC = 40\text{m}, CD = 60\text{m}$ and $CE = 75\text{m}$.	Given (diagram)
II) $\angle DCE = \angle BCA$	Vert. $\angle$'s Thm
III) $\frac{AC}{CE} = \frac{50}{75}$ and $\frac{BC}{CD} = \frac{40}{60}$	Subs. Prop of =.
IV) $\frac{AC}{CE} = \frac{2}{3}$ and $\frac{BC}{CD} = \frac{2}{3}$	Simplification.
V) $\frac{AC}{CE} = \frac{BC}{CD}$	Trans. Prop of =.
VI) $\triangle ABC \sim \triangle EDC$	SAS

⑫ Statement	Reason
I) $AB = 12, BC = 15, AE = 12, ED = 4,$ $EC = 3,$ and $DC = 5$.	Given (diagram)
II) $AD = AE + ED$ and $BD = BC + CD$	Seg Add Post.
III) $AD = 12 + 4$ and $BD = 15 + 5$	Subs. Prop of =.
IV) $AD = 16$ and $BD = 20$	Simplification.

V) $\frac{ED}{DA} = \frac{4}{16}, \frac{DC}{DB} = \frac{5}{20},$ and $\frac{EC}{AB} = \frac{3}{12}$	Subs. Prop of =.
VI) $\frac{ED}{DA} = \frac{1}{4}, \frac{DC}{DB} = \frac{1}{4},$ and $\frac{EC}{AB} = \frac{1}{4}$	Simplification.
VII) $\frac{ED}{DA} = \frac{DC}{DB} = \frac{EC}{AB}$	Trans. Prop of =.
VIII) $\triangle ABD \sim \triangle ECD$	SSS

⑬ Statement	Reason
I) $\overline{PQ} \parallel \overline{MN}$	Given (diagram)
II) $\angle OPQ \cong \angle OMN$ and $\angle OQP \cong \angle ONM$	CAP
III) $\triangle POQ \sim \triangle MON$	AA

* You may choose to use the Reflexive property of congruence.
 $\angle POQ \cong \angle POQ$